

WELCOME to ASEN 5090

Introduction to GPS/GNSS

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SpaceX launches its Falcon 9 rocket on May 30, 2025 at 1:37 p.m. EDT (1737 UTC) with the GPS III-7 Space Vehicle 08 satellite from Space Launch Complex 40 (SLC-40) at Cape Canaveral Space Force Station. Image: Michael Cain/Spaceflight Now



University of Colorado
Boulder

ASEN 5090 Fall 2026

Lockheed Martin GPS-III Model @Smead Aerospace



ASEN 5090 INTRODUCTION TO GNSS

MW 1:00-2:15PM, AERO N240

Lecture recordings available on CANVAS

Instructor

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Please include **ASEN5090** at the start of subject line in all emails

Office hours to be scheduled in-person and on zoom

– please fill out CANVAS survey



COURSE DESCRIPTION

- Core AES course: Astrodynamics and Satellite Navigation Focus Area
- Overview of satellite navigation systems with emphasis on GPS
- Fundamentals of GNSS hardware and algorithms/software
- Prerequisite for:
 - ASEN 6092 GNSS Remote Sensing
 - ASEN 6091 GNSS Receiver Architecture
 - ASEN 6090 GNSS Software and Applications



PREREQUISITES AND ELIGIBILITY

- Open to graduate students in Engineering, Physics, Applied Math, Geological Sciences, Geography, and related fields.
- Undergraduates must get instructor permission.
- Students are expected to have:
 - good problem solving skills
 - college physics, calculus, vector and matrix math, linear algebra
 - computer programming (MATLAB, Python, or C/C++)



Textbooks

Required text:

Global Positioning System – Signals, Measurements, and Performance, Pratap Misra and Per Enge, 2nd Edition or Revised 2nd Edition.

- Can be purchased at: <http://www.gpstextbook.com/>

Position, Navigation, and Timing Technologies in the 21st Century, Ed by Y.T.J. Morton, et al.

Available for download through the CU Libraries:

- Vol 1: <https://ieeexplore-ieee-org.colorado.idm.oclc.org/book/9304973>
- Vol 2: <https://ieeexplore-ieee-org.colorado.idm.oclc.org/book/9304974>



Other useful reference books

- Springer Handbook of Global Navigation Satellite Systems, Teunissen & Montenbruck eds., ISBN 978-3-319-42928-1, 2017.
- Understanding GPS - Principles and Applications, Kaplan & Hegarty eds., 3rd Edition, ISBN 978-1-630810580, 2017.
- Engineering Satellite-Based Navigation and Timing: Global Navigation Satellite Systems, Signals, and Receivers, J. Betz, ISBN 978-1-118-61597-3, 2015.
- GPS Satellite Surveying, A. Leick, 3rd Edition, ISBN 0-471-05930-7, 2014.
- GNSS Applications and Methods, Gleason & Gebre-Egziabher eds., ISBN 978-1-59693-329-3, 2009.
- Global Positioning System, Theory & Practice, Hofmann-Wellenhof, Lichtenegger, & Collins, 5th Edition, ISBN 3-211-83534-2, 2001.
- A Software-Defined GPS & Galileo Receiver, Borre et al., ISBN 0-8176-4390-7, 2007.



Magazines and Journals

- Inside GNSS: <http://www.insidegnss.com/>
- GPS World: <http://gpsworld.com/>
- GPS Solutions: <https://link.springer.com/journal/10291>
- Navigation, Journal of the Institute of Navigation: www.ion.org
- Proceedings of ION Conferences: www.ion.org
- IEEE Transactions on Aerospace and Electronic Systems:
<http://ieeexplore.ieee.org/xpl/RecentIssue.jsp?reload=true&punumber=7>

Web resources

- <http://www.gps.gov/>
- Wikipedia (really! Has good compilation of launches & satellites)

You can also find lots of hype and incorrect stuff about GPS on the web....

1-2 participation points given for sharing an excellent or problematic source of GNSS information not included here.



SUBJECT OUTLINE

- GNSS Basics
- Measurements and Errors
- Position Solutions
- GPS Signals and Receivers
- Applications



Week	Date	Topic	Reading	Assignments - Due 11pm*
1	8/24/26 8/26/26 8/28/26	L1. Introduction & GPS Basics L2. Navigation Measurements & Methods	PNT Ch 1, ME 2.1 & 2.2 ME Ch 1	HW1 Assigned
2	8/31/26 9/2/26 9/4/26	L3. GPS Signal Structure L4. Coordinate Frames	ME Ch 2, PNT Ch 2 ME Ch 4.1, 4.A	HW1 Due, HW2 Assigned
3	9/7/26 9/9/26 9/11/26	Labor Day (no class) L5. Orbits & Visibility	ME Ch 4.3	HW3 Assigned
4	9/14/26 9/16/26 9/18/26	L6. Measurement Model & IGS Data L7. Time Systems, Clocks, & Relativity	ME 5.1, 5.2, https://igs.org/ , PNT Ch 27.2 ME 4.2, Ashby 2003	HW2 Due, HW4 Assigned
5	9/21/26 9/23/26 9/25/26	L8. Ionospheric Effects & Models L9. Tropospheric Effects and Models	ME Ch 5.3.2, PNT Ch 31 ME Ch 5.3.3	HW3 Due
6	9/28/26 9/30/26 10/2/26	L10. Multipath L11. Error Mitigation & Summary	ME Ch 5.4.2+ PNT Ch 22 ME 5.6, 5.8	HW4 Due, HW5 Assigned
7	10/5/26 10/7/26 10/9/26	L12. Least Squares Solutions L13. GPS Point Solutions	ME Ch 6.A ME Ch 6.1	
8	10/12/26 10/14/26 10/16/26	L14. Improved positioning methods L15. Precise positioning	ME Ch 7	HW5 Due, HW6 Assigned
9	10/19/26 10/21/26 10/23/26	L16. Midterm Review Midterm Exam (no lecture)		Midterm Release 10/21 8AM Midterm Close 10/25 11PM
10	10/26/26 10/28/26 10/30/26	L17. Introduction to Signals & Spectra L18. GNSS Spectrum	ME Ch 8 ME Ch 9	HW6 Due, HW7 Assigned
11	11/2/26 11/4/26 11/6/26	L19. Receiver Overview, Antennas, Link Budget L20. GNSS Receiver Architecture	ME Ch 11, PNT Ch 14 ME Ch 10 (parts)	
12	11/9/26 11/11/26 11/13/26	L21. GPS Acquisition L22. GPS Acquisition (continued)	ME Ch 11.2	HW7 Due, HW8 Assigned
13	11/16/26 11/18/26 11/20/26	L23. GPS Signal Tracking Basics L24. Advanced Signal Tracking (Xu)	ME Ch 12, PNT Ch 15	
14	11/23/26 11/25/26 11/27/26	FALL BREAK		
15	11/30/26 12/2/26 12/4/26	L25. Space Applications of GNSS L26. Aircraft and UAV Applications L27. Final Exam Review (Monday Class Schedule)	PNT Ch 62-63 PNT Ch 60-61	HW8 Due
16	12/7/26 12/9/26 12/11/26	Final Exam available starting Friday, 12/4 Final Exam due Tues, 12/8 (24hr time limit)		Final Release 12/4 6PM Final Close 12/8 11PM

Midterm Exam - 24-hr take home exam

Final Exam - 24-hr take home exam

Students are permitted up to two late HW submissions by the Sunday after the Friday deadline at 11pm without penalty.

Homeworks are due on Fridays at 11pm.

- Students may submit up to two late submissions by the following Sunday at 11pm without penalty.
- Lowest homework score (out of 8) is dropped

Take home exams:

- Midterm Exam Oct 21-25 (choose a 24-hr period)
- Final Exam, Dec 4-8 (choose a 24-hr period)



ASSIGNMENTS

- 8 Homework/Project Assignments
- Problems from the book
- Developing your own code for GPS solutions and acquiring signals
- Collaboration is permitted on these assignments. This means you may discuss the means and methods for solving problems and even compare answers, but you are not free to copy solutions from classmates or from internet resources.
Any figures or other materials used/copied from other sources must include an appropriate citation.
- Submissions uploaded to Gradescope by specified deadline.



Grading Policies

Participation	5%
Project assignments	70% Average of 7 highest HW scores
Midterm Exam	10%
<u>Final Exam</u>	<u>15%</u>
Total	100%

Participation includes in-class, office hours, online discussions, & other opportunities.



Conditional use of AI in this class

- You are permitted (but not required) to use generative AI tools in this course for the following purposes only:
 - * partial code generation or code verification (i.e. $\leq 30\%$ as measured by total lines of code submitted) for programming on homework assignments;
 - * to review or study course material and topics on your own.
- If you use gen AI tools for partial code generation or verification on homework, you must document your usage by stating this explicitly at the top of your assignment and cite the portions (lines of code) that were generated or checked by the specific tool you used. Failure to properly document gen AI usage for coding assignments on homework will be treated as a potential CU Honor Code violation.
- **You may not use gen AI tools to produce solutions for other nonprogramming questions on homework.**
- **You may not use gen AI tools for any kind of question (whether programming or non-programming) on the midterm or final exams.** Exam work submitted for grading must conform to these guidelines, in order to receive full credit reflective of your own individual competency, learning, and understanding of the material.



University Policies – See Syllabus for Full Descriptions

- Honor Code (Do not plagiarize or cheat on any assignment in this class.)
- Accommodation for Disabilities (Based on letters from DS)
- Religious Obligations (Let instructors know at least 2 weeks before exam)
- Preferred Student Names and Pronouns (Please record on Name Coach)
- Classroom Behavior (Be polite and respectful to all, in-person and online)
- Sexual Misconduct, Discrimination, Harassment and/or Related Retaliation
- Mental Health and Wellness - [Counseling and Psychiatric Services \(CAPS\)](#), is here to help. CAPS offers counseling, referrals, psychiatric care, crisis support, and much more. Visit CAPS in the C4C, or call (303) 492-2277, 24/7.



University Policies - Sexual Misconduct, Discrimination, Harassment and/or Related Retaliation

- CU Boulder is committed to fostering a productive and welcoming learning, working, and living environment. University policy prohibits [protected-class](#) discrimination and harassment, sexual misconduct (harassment, exploitation, and assault), intimate partner abuse (dating or domestic violence), stalking, and related retaliation by or against members of our community on- or off-campus. Denial of an approved accommodation for disability, religious observance, or pregnancy or pregnancy related medical conditions may be discriminatory.
- The [Office of Institutional Equity and Compliance \(OIEC\)](#) addresses these concerns, and individuals who have been subjected to misconduct can contact OIEC at 303-492-2127 or email OIEC@colorado.edu. Information about university policies, [OIEC reporting options](#), and [OIEC support resources](#) including confidential services can be found on the [OIEC website](#).
- Faculty and graduate instructors are required to inform OIEC when someone discloses misconduct regardless of when or where it occurred. This is to ensure that those impacted receive outreach from OIEC about resolution options and support resources. To learn more about reporting and support options for a variety of concerns, visit the [Don't Ignore It page](#).



GPS Basics

Outline for today

- Space Segment
- Control Segment
- User Segment
- Position solution – ranging and geometry

Questions to think about:

- How does a receiver know where the satellites are?
- Where are they?
- What is needed to find position with GPS?
- Reading – PNT Ch1 & ME Ch 2.1 & 2.2

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	8/28/26		HW1 Assigned
2	8/31/26	L3. GPS Signal Structure	
	9/2/26	L4. Coordinate Frames	
	9/4/26		HW1 Due, HW2 Assigned



What's so special about GNSS (that it deserves its own class)?

- It actually is really amazing...
- Unprecedented accuracy – “stand alone” position good to few meters, differential good to $< 1\text{m}$, “survey” at the mm-cm
- Global availability – all weather, all the time, everywhere, seamlessly
- Global presence – today there are 4 independent GNSS's + many augmentations
- Applications – land, marine, airborne, satellites, time transfer – you name it.
- Highly multidisciplinary
 - to fully understand it: how it works, what can it do, how is it vulnerable, requires putting together knowledge from many different fields.
- Cost – system is paid for entirely by government (i.e. taxpayers).
 - No fees to manufacturers or users
 - Receivers can be produced for $\ll \$100$

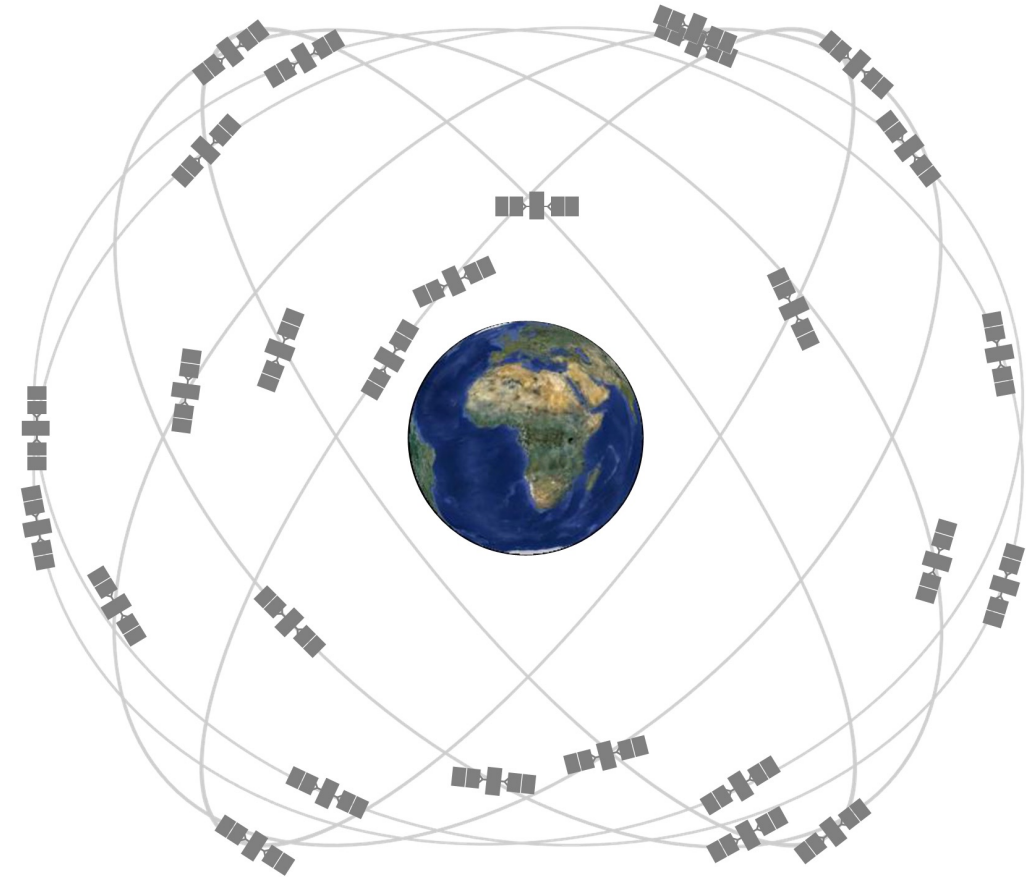


GLOBAL NAVIGATION SATELLITE SYSTEMS (GNSS)

- US Global Positioning System
- Russian GLONASS
- European Galileo
- Chinese BeiDou
- + other augmentations



Constellation of satellites that transmit continuous
to allow 1-way ranging by passive users
Primarily in medium Earth orbits (MEO)



GPS Constellation



KEY REQUIREMENTS

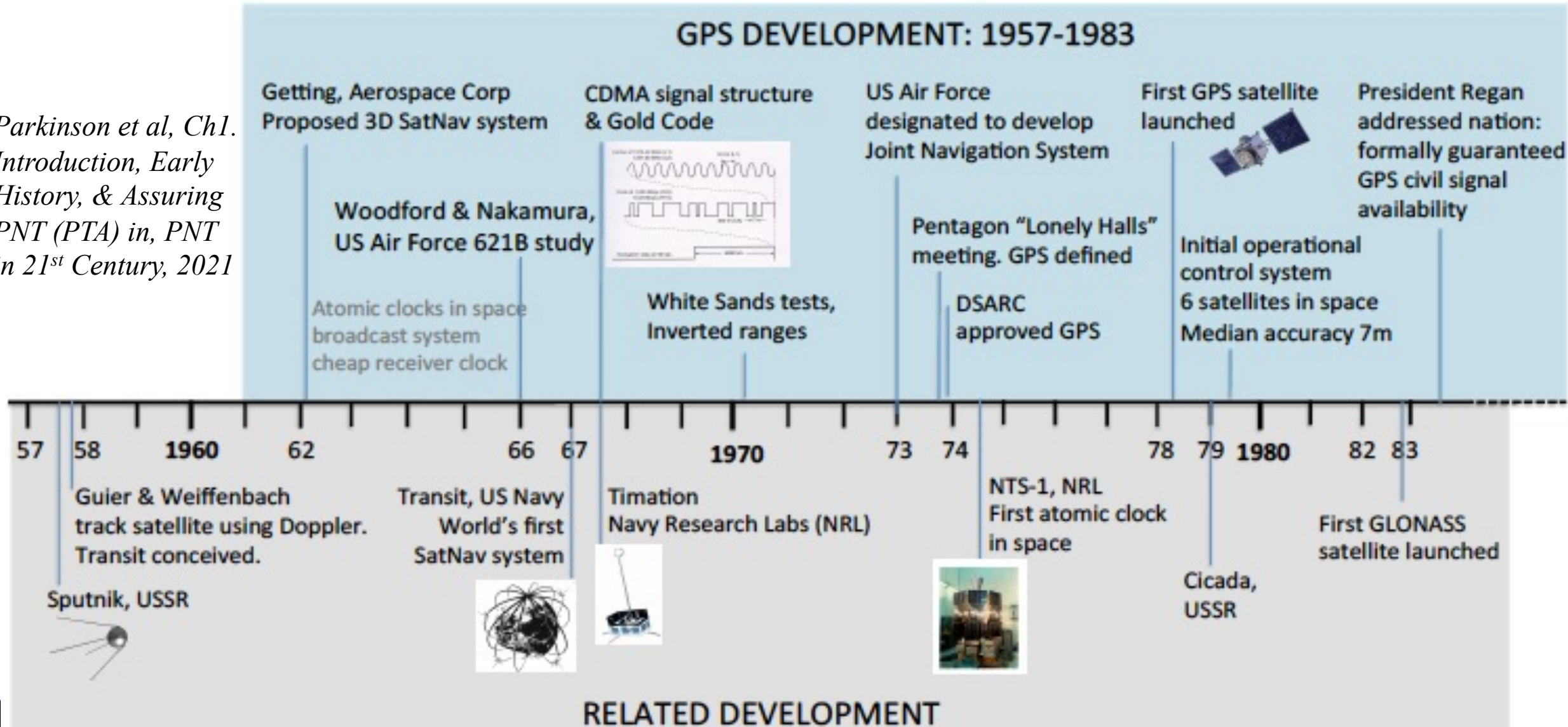
- High accuracy, precision
- 24/7 operation, all weather, day & night
- Dual use – both military and civil users
- Self contained and passive for the user

“The United States Government shall... Provide on a continuous, worldwide basis civil space-based positioning, navigation, and timing services free of direct user fees for civil, commercial, and scientific uses and for homeland security through the Global Positioning System and its augmentations, and provide open, free access to information necessary to develop and build equipment to use these services.”

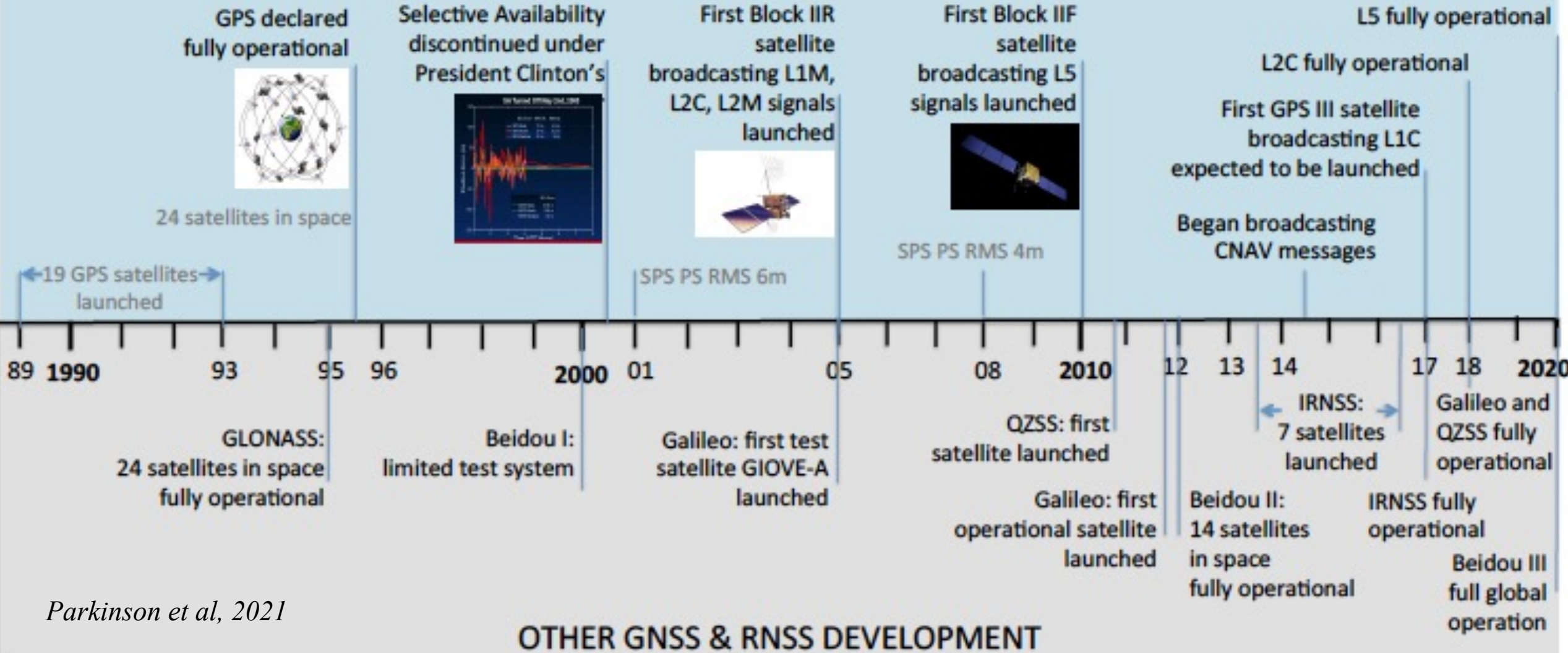
President George W. Bush, U.S. National Space-Based Positioning, Navigation, and Timing Policy, December 2004

GPS History: Early Development

*Parkinson et al, Ch1.
Introduction, Early
History, & Assuring
PNT (PTA) in, PNT
in 21st Century, 2021*



GPS DEVELOPMENT: 1989-2020



Parkinson et al, 2021



DISCIPLINES/SUBJECT AREAS RELEVANT to GNSS

- Ranging Signals:
 - Electromagnetics, signal processing, coding, antennas, jamming & interference
- Satellites:
 - Orbits, relativity, perturbations, attitude control, launch/maintenance, atomic clocks
- Environment
 - Ionosphere, troposphere, antenna installation
- Users
 - Platform dynamics, challenging environments (antenna shading), computational capability, estimation methods, accuracy requirements (aided or stand-alone)
- Politics
 - Spectrum usage, privacy, fees, security, national pride...

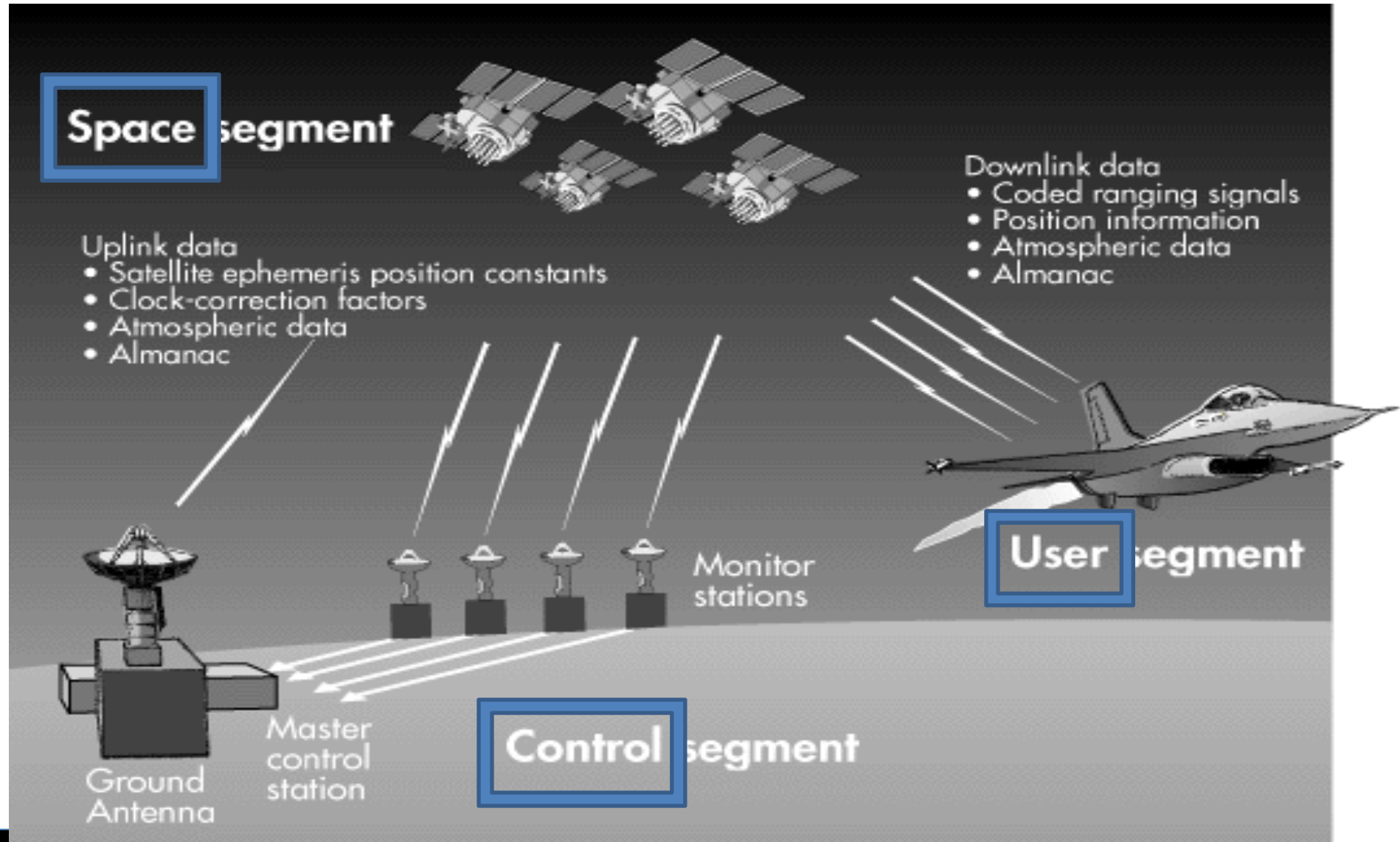


RISKS TO GPS/GNSS OPERATION

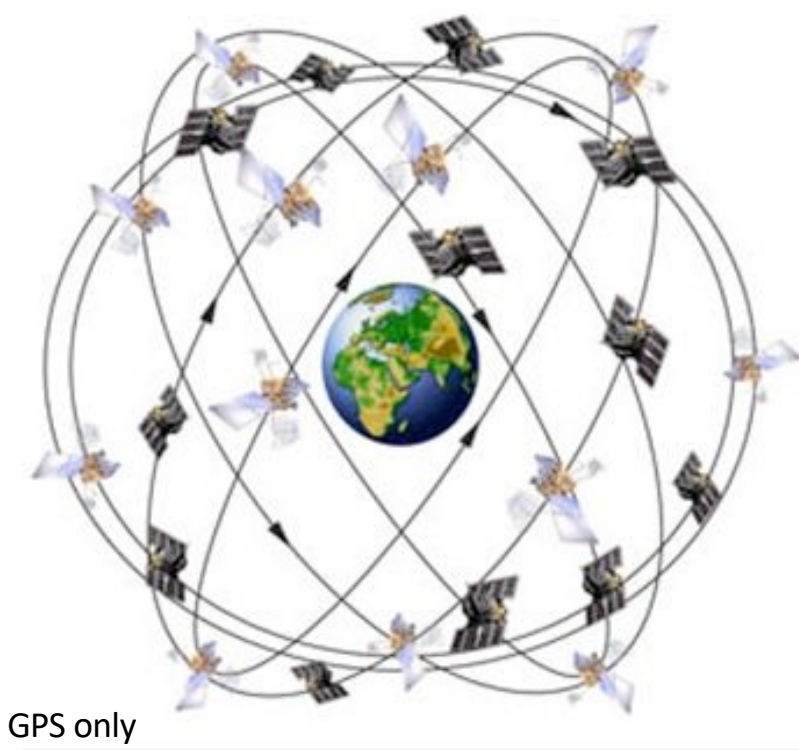
- Local interference – intentional (jamming & spoofing) or unintentional
- Space weather effects – rapid variations in ionospheric activity
- Incorrect/misleading signals or nav messages – system glitch or intentional spoofing
- Control system issues – as experienced by GLONASS and Galileo systems
- User equipment issues – week rollovers, code not fully compliant with specs
- Loss of satellites



Classic GPS



GNSS Constellations



~140 Active GNSS Satellites

- GPS (US) – 32 MEO
- GLONASS (Russia) – 26 MEO
- Galileo (EU) – 27 MEO
- BeiDou (China) – 47 MEO/GEO/IGSO
- QZSS (Japan) – 1 GEO/ 3 IGSO (total of 7 planned)
- IRNSS (India) aka NavIC - 3 GEO/ 4 IGSO
- SBAS (various) GEO to support GNSS

Medium Earth Orbit (MEO) – orbital periods ~ 12 hours

Geostationary Orbits (GEO) – orbital period of 1 day, 0 deg inclination

Inclined Geosynchronous Orbits (IGSO) – orbital period of 1 day, inclined

Space Segment (1): GPS Constellation

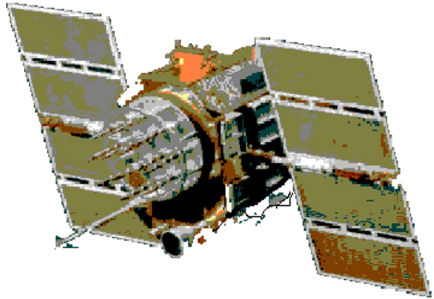


- Baseline constellation: 24 satellites
- Current active satellites: 32
- 6 orbital planes
- 4~6 satellites per plane
- 55° inclination angle
- near circular orbit
- 26,560 km orbit radius
- ~12 hours orbital period
(~ $\frac{1}{2}$ sidereal day 11 hour 58 min)

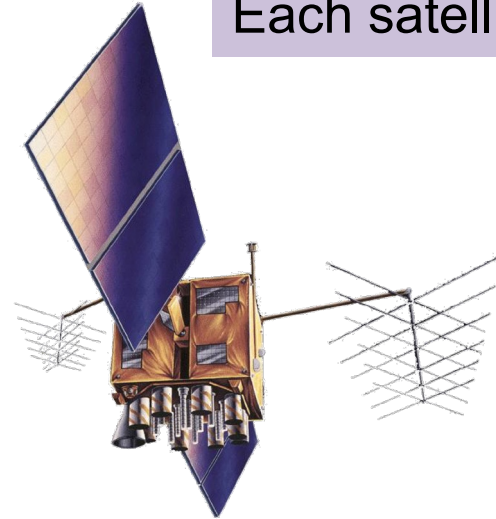


GPS Satellite Blocks & Status (July 2026)

31 active, 1 commissioning, 1 residual



Block II/IIA
Boeing (Rockwell)
Launched 1989 – 1997



Transmit antenna array points toward the Earth
Each satellite carries 3-4 atomic clocks

Block IIR/IIR-M
Lockheed Martin
Launched 1997 – 2009
11 Operational, 1 possibly in residual status



Block IIF
Boeing
Launched 2010-2016
11 Operational

GPS III
Lockheed Martin
Launched 2018 -
9 Operational, 1 commissioning



Check out their
nicknames!
SVN79 is *Amelia Earhart*
SVN81 is *Katherine Johnson*



GPS Jargon

- **PRN (pseudorandom noise code)** tells you which code is being generated by a certain satellite.
The codes get reused because for legacy system there are only 32 available for assignment to satellites
- **SVN (satellite vehicle number)** tells you exactly which spacecraft it is.
This number is not reused as it relates to the design specifications of the satellite.
- Constellation status is maintained by the Coast Guard, (NANU)
<https://www.navcen.uscg.gov/?Do=constellationStatus>
- For example, PRN 10 was assigned to
 - SVN40 from 1996-07-16 00:00:00 to 2015-08-03 23:59:59
 - SVN36 from 2015-09-16 00:00:00 to 2015-10-26 23:59:59
 - SVN73 from 2015-10-31 00:00:00 – present
- Mapping of PRNs to SVNs is available here: <https://celestrak.org/GPS/GPSData.txt>



Constellation Status

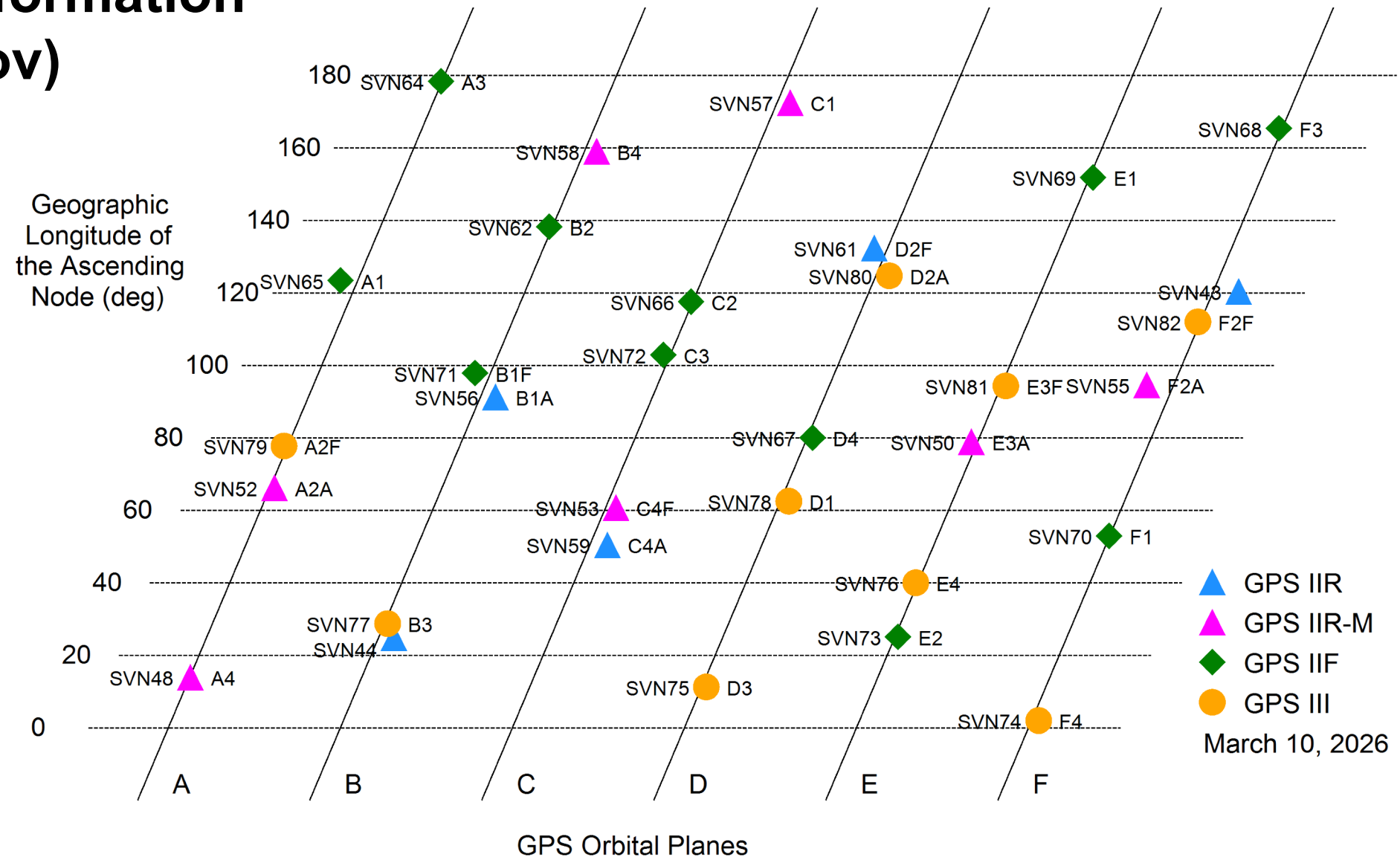
(navcen.uscg.gov)

GPS CONSTELLATION STATUS FOR Sun Aug 23 2026								
Plane	Slot	SVN	PRN ▲	Block-Type	Clock	Outage Start	NANU Type	NANU Subject
D	2	80	1	III	RB			
D	1	61	2	IIR	RB			
E	1	69	3	IIF	RB			
F	4	74	4	III	RB			
E	3	50	5	IIR-M	RB			
D	4	67	6	IIF	RB			
A	4	48	7	IIR-M	RB			
C	3	72	8	IIF	RB			
F	3	68	9	IIF	RB			
E	2	73	10	IIF	RB			
D	5	78	11	III	RB			
B	4	58	12	IIR-M	RB			
A	5	43	13	IIR	RB	24 APR 202624 APR 2026	DECOM	SVN43 (PRN13) DECOMMISSIONING JDAY 114/2200

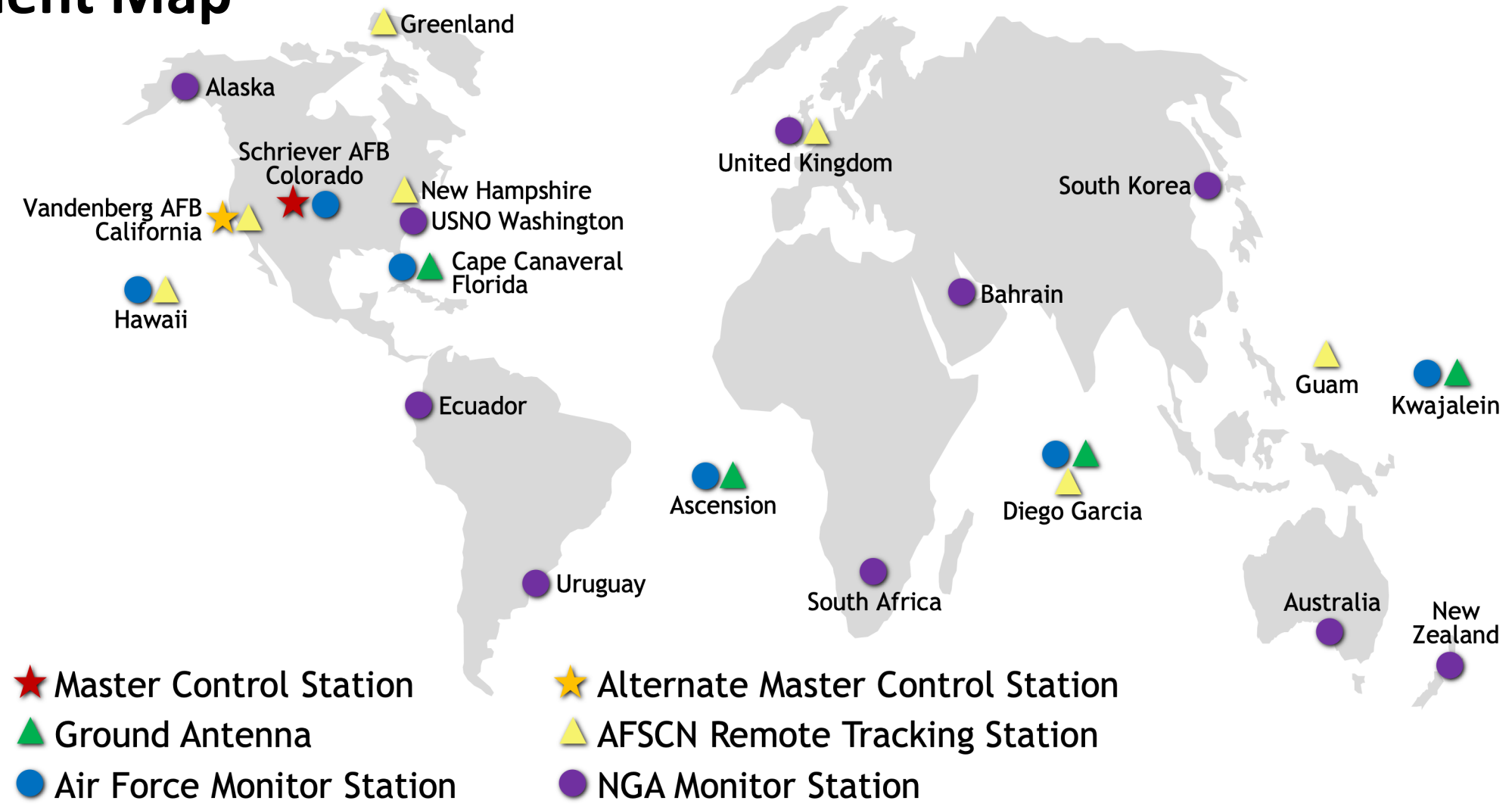


Constellation Information

(navcen.uscg.gov)



Control Segment Map



<https://www.gps.gov/multimedia/images/GPS-control-segment-map.pdf>

Control Segment Functions

- Master Control Station (MCS): Schriever AFB, CO
 - Operates other monitor stations, provides command and control functions
- Monitor Stations: unmanned, total 15 around the globe
 - 6 USAF Monitor Stations
 - 9 National Geospatial-Intelligence Agency (NGA)
 - 4 Ground antennas (GA) for command and control
 - Each SV is monitored by at least 2 stations at any time simultaneously
 - Equipped with Cs clock, meteorological instruments, communication links with MCS/SV
- Control Segment Functions:
 - Monitor SV orbit and health
 - Maintain GPS time
 - Predict SV ephemerides and clock parameters
 - Update SV navigation message
 - Command SV maneuvers to maintain orbit



User Segment: Early Receivers



Manpack,
Rockwell Collins
1978



Texas Instruments-4100, 1st widely used commercial
receiver
1981



Magellan NAV1000
1st handheld GPS
1989



Benefon!
1st commercial mobile phone
w/GPS
1999



User Segment: Modern GNSS Receivers



GnssLogger						
MapPlotsStatusSkyplotAGNSSV						
ID	GNSS	Freq	C/N0	Used	Azim	Elev
30	E1	31.4		Y	342	78
30	E5A	27.0		Y	342	78
7	L1	35.8			3	75
7	E1	32.2		Y	251	64
7	E5A	21.9			251	64
8	L5	34.9		Y	77	53
8	L1	39.5		Y	77	53
30	L5	27.5		Y	309	51
30	L1	29.4		Y	309	51
13	G1	32.6		Y	35	49
4	G1	35.2		Y	294	46
14	G1	38.9		Y	298	44
9	L1	30.0			177	40
3	G1	28.5		Y	218	40
14	L5	22.6		Y	250	35
14	L1	31.8		Y	250	35
8	E1	28.2		Y	172	34
27	E1	24.5		Y	253	34
27	E5A	21.6			253	34
23	G1	38.8		Y	99	30
14	E1	21.0			185	27
27	L5	35.2		Y	45	23
27	L1	39.9		Y	45	23
22	L1	33.8		Y	243	18
4	L5	25.4		Y	160	16
4	L1	37.7		Y	160	16



GPS Services

- Standard Positioning Service: SPS
 - Civilian users or open GPS service
- Precise Positioning Service: PPS
 - Service for authorized or military users
- Both services provide position, velocity, and time
- Performance Standard (PS): defines the performance of GPS satellite broadcast signals in terms of accuracy, integrity, continuity, and time accuracy



GPS Performance Specification

Performance	Definition	SPS Value	PPS Value
Availability	Global mean daily PDOP	≤ 6 (98%)	≤ 6 (98%)
	Worst site daily PDOP	≤ 6 (88%)	≤ 6 (98%)
Accuracy	Mean global SISRE at any AOD	12.8m (95%)	12.8m (95%) w/o WAGE 4.4m (95%) w/ WAGE
Integrity	Probability of instantaneous SISRE exceeding threshold w/o timely alert	10^{-5}	None
Time to alert	Warning when signal should not be used	10 seconds	10 seconds
Continuity	Probability of not losing signal availability in any one hour	0.9998	0.9998
Time accuracy	Mean UTC offset error at any AOD	40 ns (95%)	40 ns (95%)

PDOP: Position dilution of precision

WAGE: Wide Area GPS Enhancement

AOD: Age Of Data

SISRE: Signal In Space user REquirement

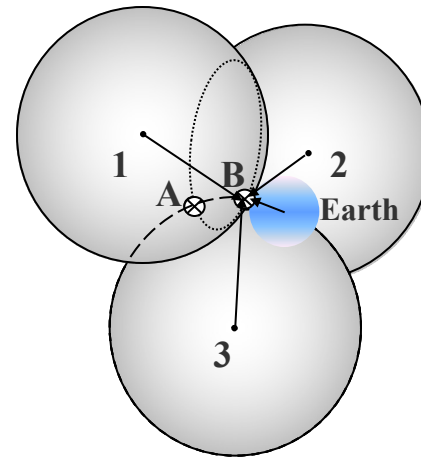
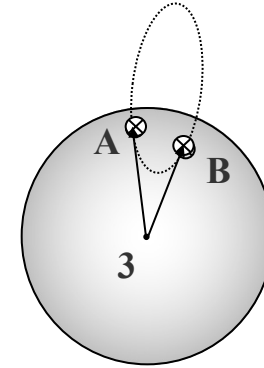
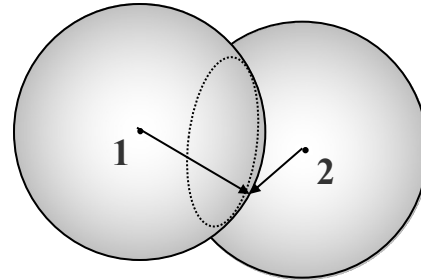
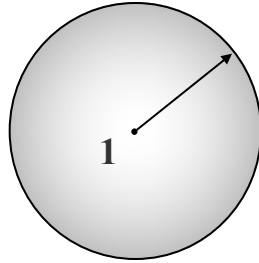


Fundamentals of Satellite Navigation

- Satellite based navigation is fundamentally based on:
 - The precise measurement of time
(you have to agree on what you mean by time!)
 - The constancy of the speed of light
- GNSS uses the concept of **trilateration**:
 - Satellite (transmitter) positions are known
 - Receiver position is unknown
 - Satellite-to-receiver range measurements are used to estimate position



TRILATERATION



Position Solution

- The position solution involves an equation with four unknowns:
 - Receiver position (x, y, z) (in what reference frame)
 - Receiver clock correction (correction to what?)
 - *Position accuracy of ~1 m implies knowledge of the receiver clock to within ~3 ns*
 - GPS accuracy is based almost entirely on knowing satellite orbits and satellite clocks.
- Requires simultaneous measurements from at least four satellites
 - The receiver makes a range measurement to the satellite by measuring the signal propagation delay
 - A data message modulated on the ranging signal provides the precise location of the satellite and corrections for the satellite clock.



Measurement Equation

- GPS receiver measures “pseudorange” by measuring the apparent transit time of the signal:

$$\rho_r^s = c(t_r - t^s)$$

time of signal reception,
(based on receiver clock,
can be significantly in error)

time of transmission,
encoded in signal by
GPS satellite clock
(known precisely)



Measurement Equation (cont)

- Measured pseudorange to a satellite is comprised of:

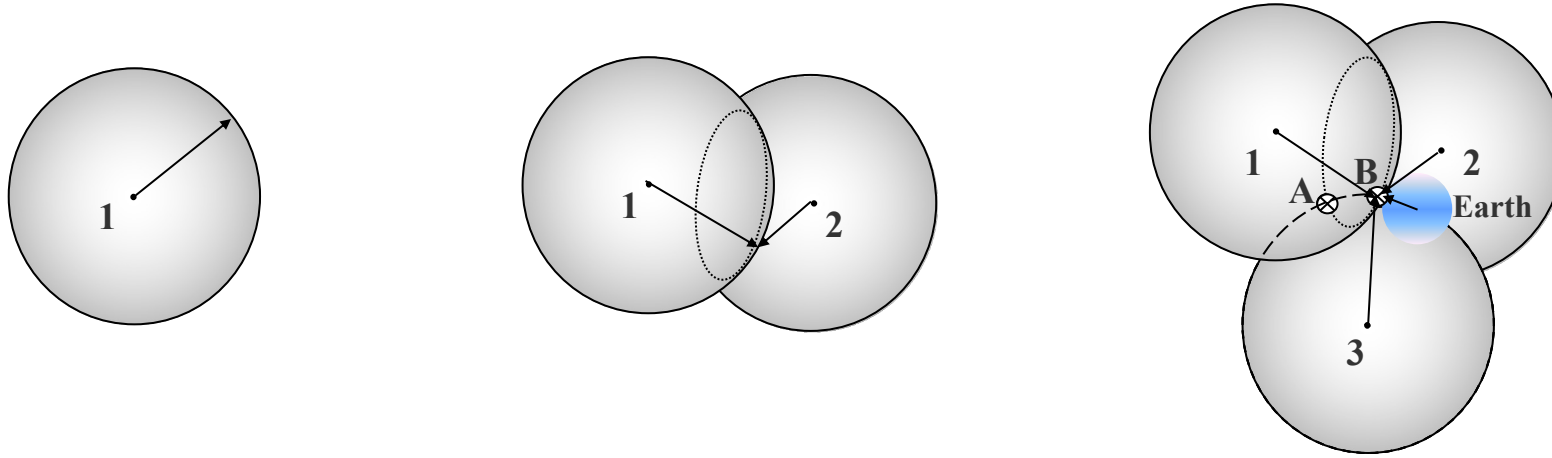
$$\rho_r^s = R_r^s - c\delta^s + c\delta_r + \varepsilon_{iono} + \varepsilon_{trop} + \varepsilon_{multi} + \varepsilon_{rel} + \varepsilon_{\rho}$$

- true range
- receiver clock error
- satellite clock error (known)
- ionosphere and troposphere delays (estimated or measured)
- other errors (satellite ephemeris and clock mis-modeling, measurement errors, multipath, receiver noise)

$$R_r^s(t_r) = \left| \vec{X}^s(t^s) - \vec{X}_r(t_r) \right|$$

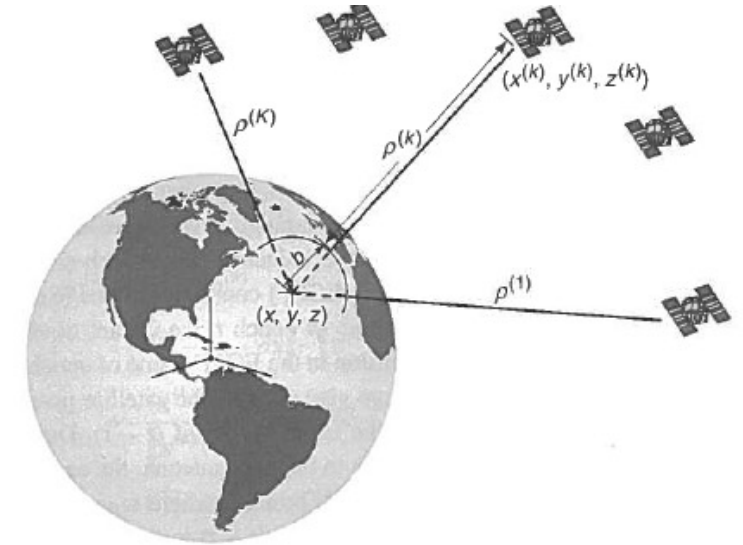


Use 4 or more pseudorange measurements to solve for user position and clock



$$\rho = R + I_\rho + T_\rho + c(\delta t_u - \delta t^s) + \varepsilon_\rho$$

trilateration



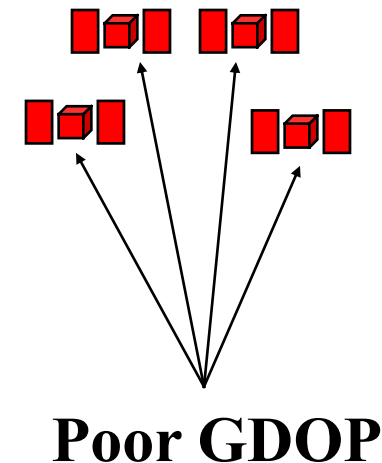
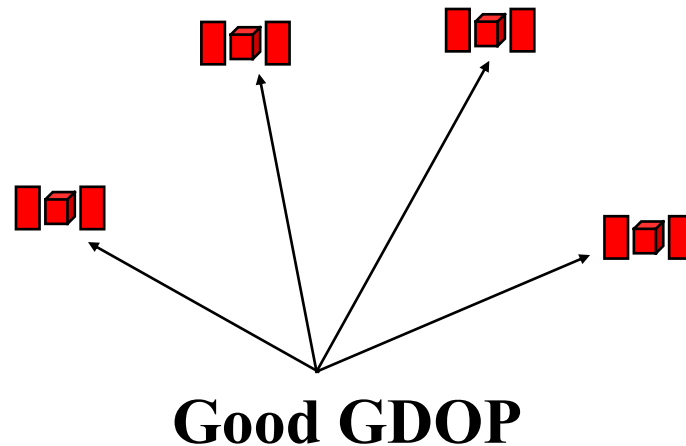
Solution Accuracy

- Two primary factors affect the fundamental position and time accuracy possible from the system:
 - **Ranging error** – a function of the quality of the signal, knowledge of the transmitter location and clock bias, environmental effects, receiver design
 - **Geometry** – the distribution of satellites in the sky

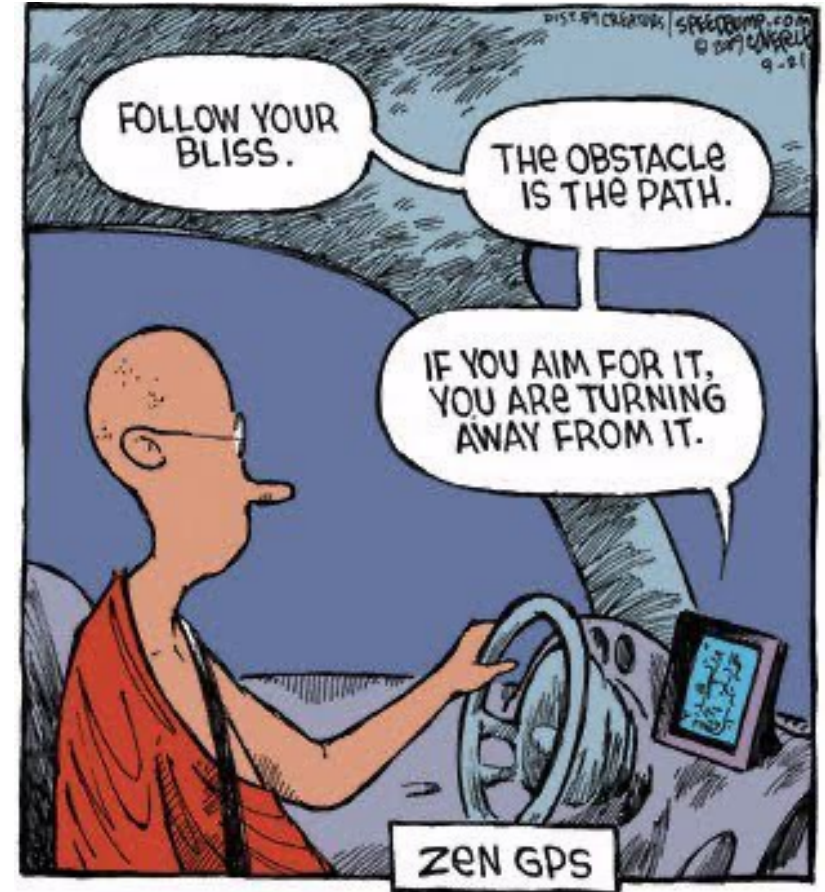


Geometry – Dilution of Precision

- Geometric Dilution of Precision (GDOP) is a measure of the quality of the receiver-to-GPS satellite range geometry
 - DOPs exist for position, horizontal, vertical, and time dilutions of precision
- Used in conjunction with the URE to forecast navigation and timing performance, weight measurements



What is your interest in GPS/GNSS?



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